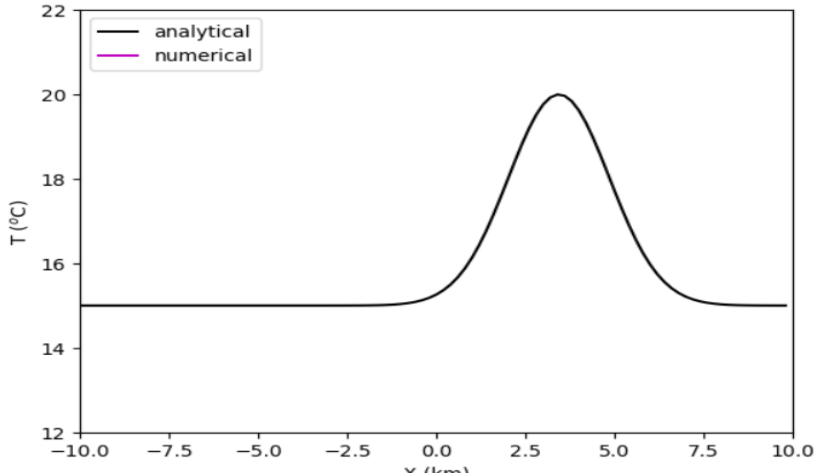
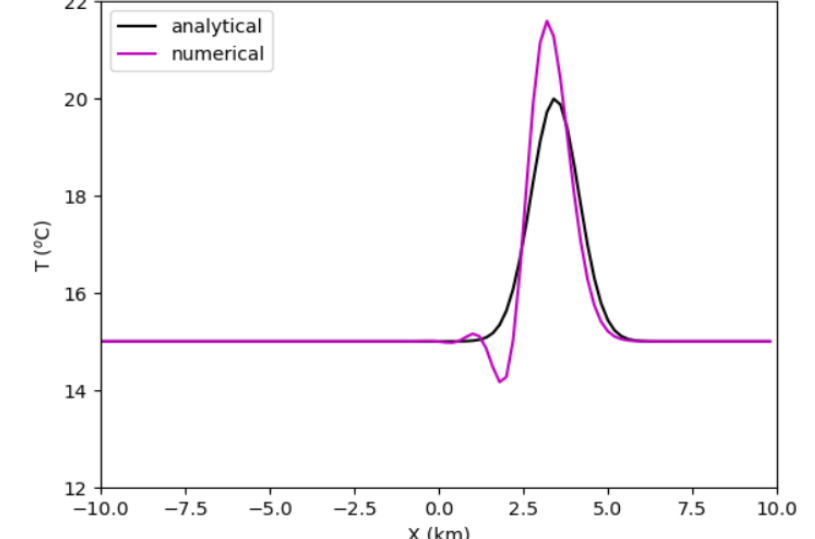
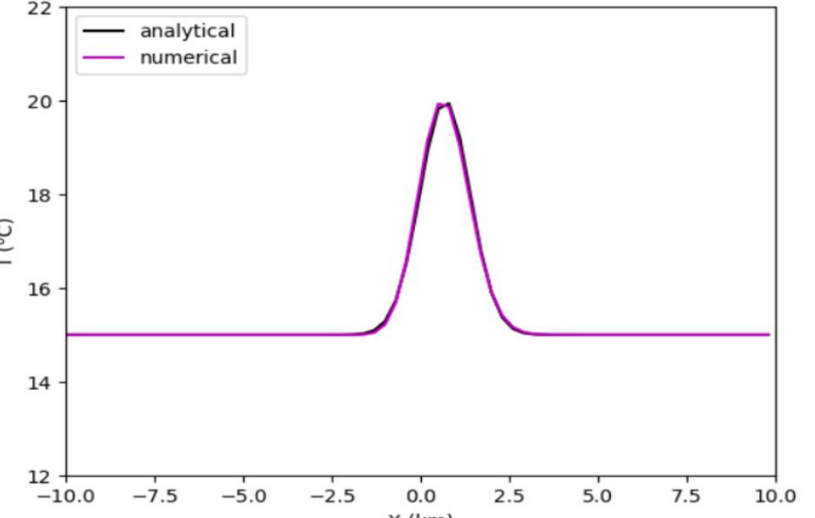
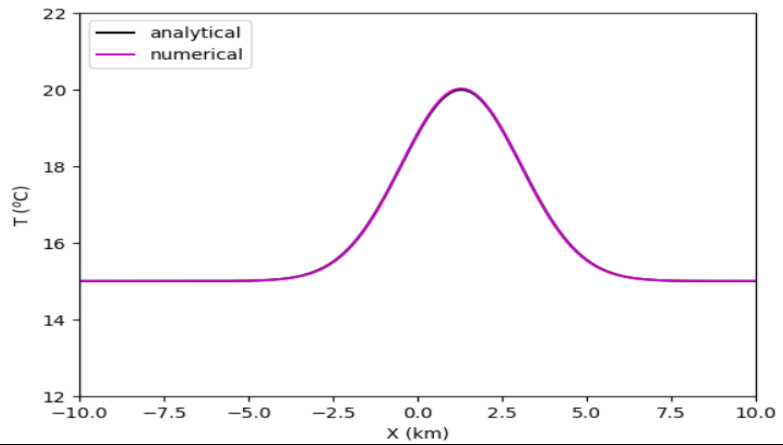


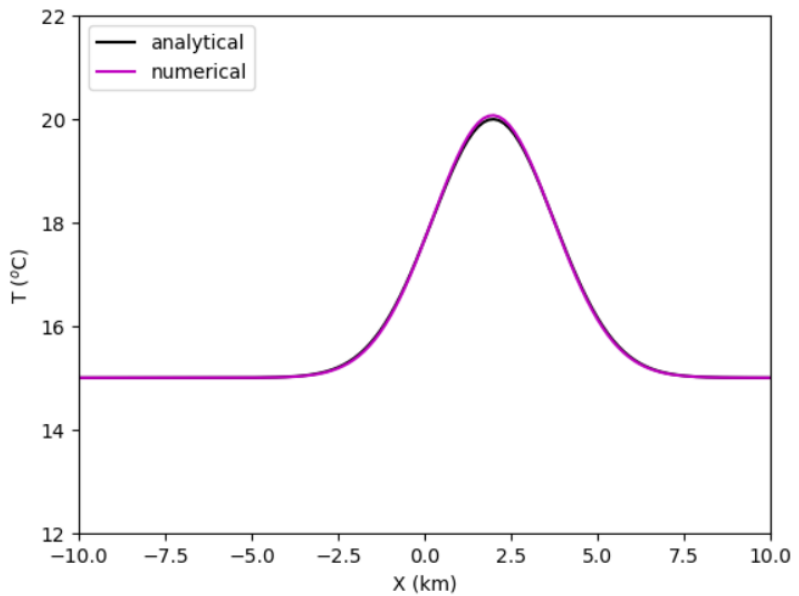
Exercise 10

Space-time parameters	Plots results
Grid steps : 100 Number of time steps : 87 Timestep = 200 s Max numerical speed $DX/DT = 1.0$ m/s Physical speed = 0.2 m/s Courant number = 0.2	 The plot shows a smooth, symmetric bell-shaped curve centered at $X = 3.5$ km, with a peak temperature of approximately 20.5°C. The analytical solution (black line) and numerical solution (magenta line) are nearly identical.
Grid steps : 100 Number of time steps : 44 Timestep = 200 s Max numerical speed $DX/DT = 1.0$ m/s Physical speed = 0.4 m/s Courant number = 0.4	 The plot shows a sharp, asymmetric peak centered at $X = 3.5$ km, with a peak temperature of approximately 20.5°C. The numerical solution (magenta line) shows significant oscillations and a small dip around $X = 1.5$ km, while the analytical solution (black line) is smooth.
Grid steps : 67 Number of time steps : 87 Timestep = 100 s Max numerical speed $DX/DT = 3.0$ m/s Physical speed = 0.08 m/s Courant number = 0.02666666666666667	 The plot shows a sharp, symmetric peak centered at $X = 1.5$ km, with a peak temperature of approximately 20.5°C. The numerical solution (magenta line) is nearly identical to the analytical solution (black line).

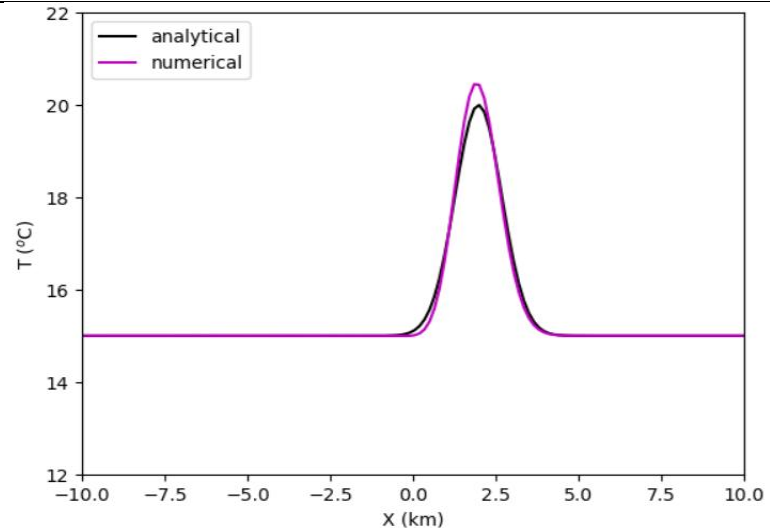
Grid steps : 134
 Number of time steps : 44
 Timestep = 200 s
 Max numerical speed
 $DX/DT = 0.75 \text{ m/s}$
 Physical speed = 0.15 m/s
 Courant number = 0.2



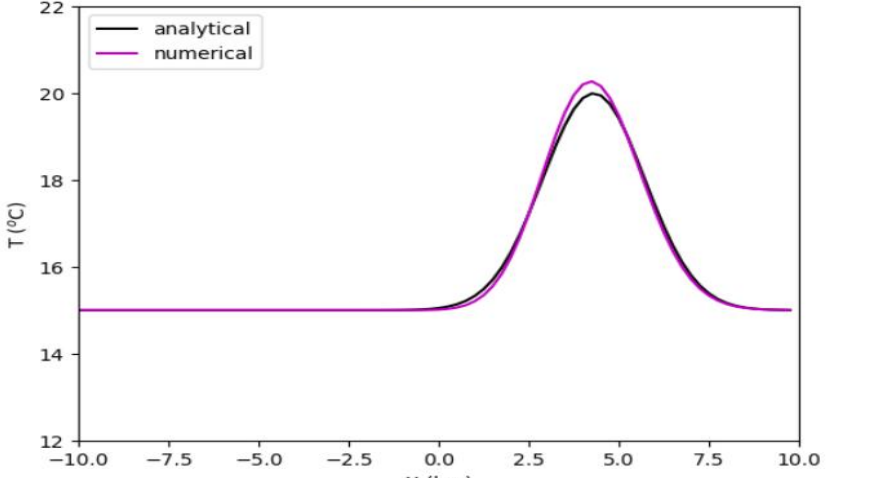
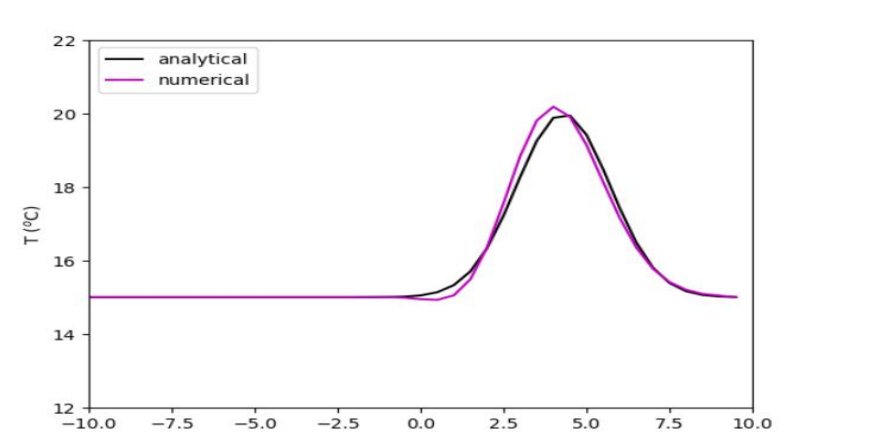
Grid steps : 134
 Number of time steps : 44
 Timestep = 200 s
 Max numerical speed
 $DX/DT = 0.75 \text{ m/s}$
 Physical speed = 0.23 m/s
 Courant number =
 0.30666666666666664
 Increased the parameter σ :
 2500



decrease the parameter σ :
 1000
 Grid steps : 134
 Number of time steps : 44
 Timestep = 200 s
 Max numerical speed
 $DX/DT = 0.75 \text{ m/s}$
 Physical speed = 0.23 m/s
 Courant number =
 0.30666666666666664

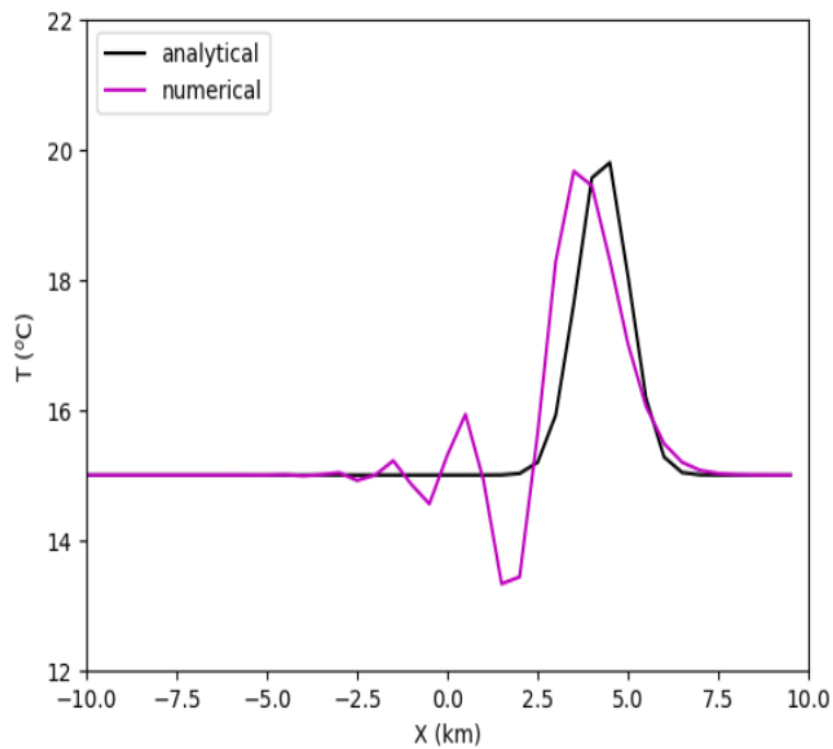


Exercise 11- question 5

Space-time parameters	Plots results
Grid steps : 80 Number of time steps : 87 Timestep = 100 s Max numerical speed $DX/DT = 2.5 \text{ m/s}$ Physical speed = 0.5 m/s Courant number = 0.2	 The plot shows the temperature distribution T in degrees Celsius as a function of position X in kilometers. The x-axis ranges from -10.0 to 10.0 km with major ticks every 2.5 km. The y-axis ranges from 12 to 22 °C with major ticks every 2 °C. A bell-shaped curve is centered at $X = 5.0$ km, with a peak temperature of approximately 20.5 °C. The curve starts at a baseline of 15 °C for $X < 0$ and returns to 15 °C for $X > 10$. The legend indicates two data series: 'analytical' (black line) and 'numerical' (magenta line). The two lines are nearly indistinguishable, indicating high accuracy of the numerical solution for this Courant number.
Grid steps : 40 Number of time steps : 87 Timestep = 100 s Max numerical speed $DX/DT = 5.0 \text{ m/s}$ Physical speed = 0.5 m/s Courant number = 0.1	 The plot shows the temperature distribution T in degrees Celsius as a function of position X in kilometers. The x-axis ranges from -10.0 to 10.0 km with major ticks every 2.5 km. The y-axis ranges from 12 to 22 °C with major ticks every 2 °C. A bell-shaped curve is centered at $X = 5.0$ km, with a peak temperature of approximately 20.5 °C. The curve starts at a baseline of 15 °C for $X < 0$ and returns to 15 °C for $X > 10$. The legend indicates two data series: 'analytical' (black line) and 'numerical' (magenta line). The two lines are nearly indistinguishable, indicating high accuracy of the numerical solution for this Courant number.

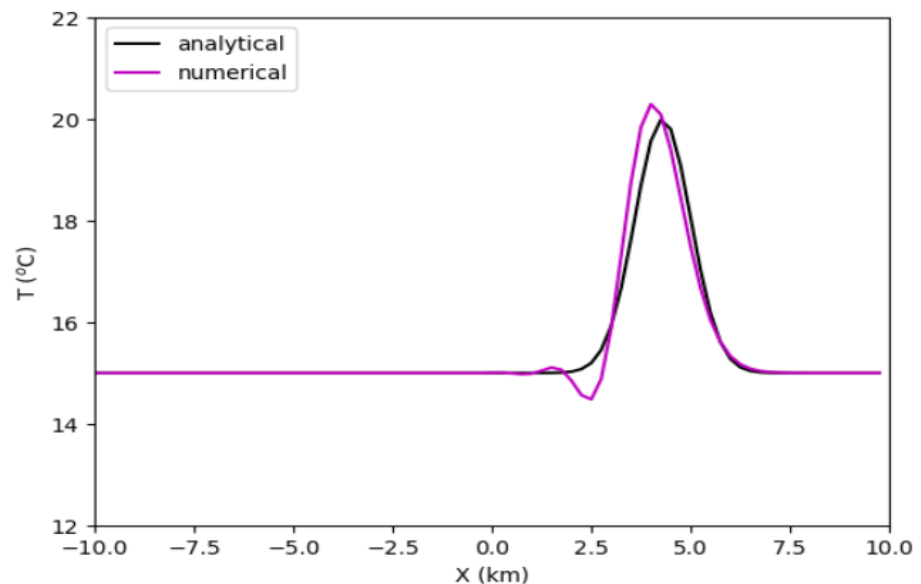
Question 6

If $c=0.5$, $dx=500$,
 $dt=100$,
 $\sigma=1000$
 Grid steps : 40
 Number of time
 steps : 87
 Timestep = 100 s
 Max numerical
 speed $DX/DT = 5.0$
 m/s
 Physical speed =
 0.5 m/s
 Courant number =
 0.1

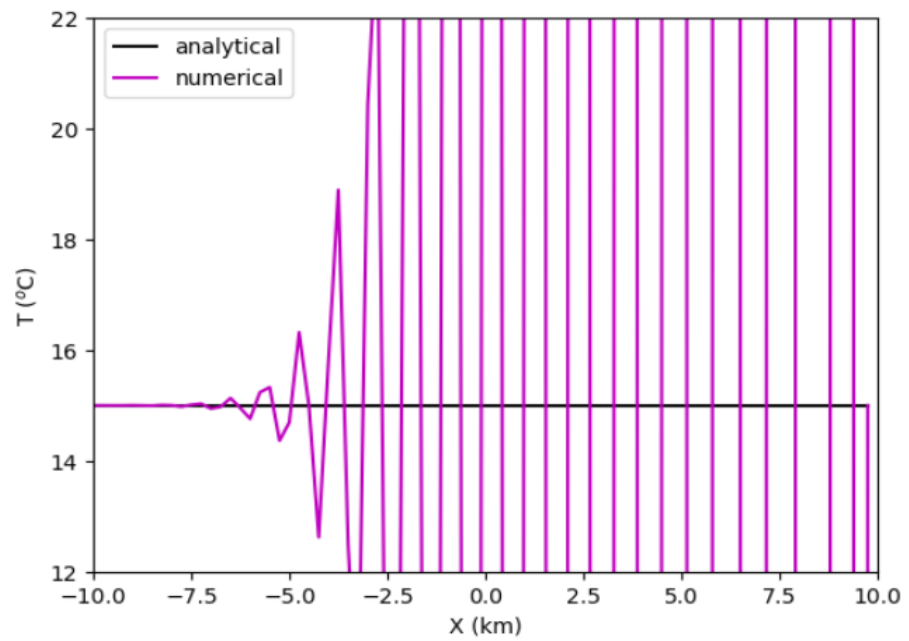


Grid steps : 80
 Number of time
 steps : 173
 Timestep = 50 s
 Max numerical
 speed $DX/DT = 5.0$
 m/s
 Physical speed =
 0.5 m/s
 Courant number =
 0.1

This plot shows when to recover the same degree of accuracy of the previous solution.

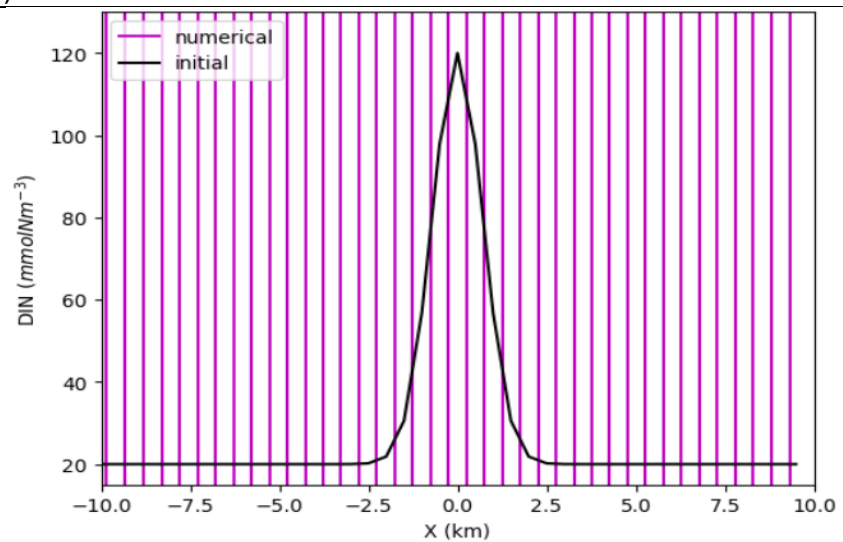


Grid steps : 80
 Number of time steps : 173
 Timestep = 50 s
 Max numerical speed $DX/DT = 5.0$ m/s
 Physical speed = 2 m/s
 Courant number = 0.4



Exercise 12 (DIFFUSION)

Grid steps : 40
 Number of time steps : 104
 Timestep = 100 s
 Max numerical diffusion $DX^2/DT = 2500.0$ m²/s
 Physical diffusion = 2000.0 m²/s
 Courant number (diffusion) = 0.8
 If $\kappa=2000$ and $dt=100$



Grid steps : 40
Number of time
steps : 208
Timestep = 50 s
Max numerical
diffusion $DX^2/DT =$
5000.0 m²/s
Physical diffusion =
2000.0 m²/s
Courant number
(diffusion) = 0.4
If dt=50

